

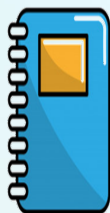
CSE 3203 – Software Engineering

Lecture 04: Soft. Eng. Ethics

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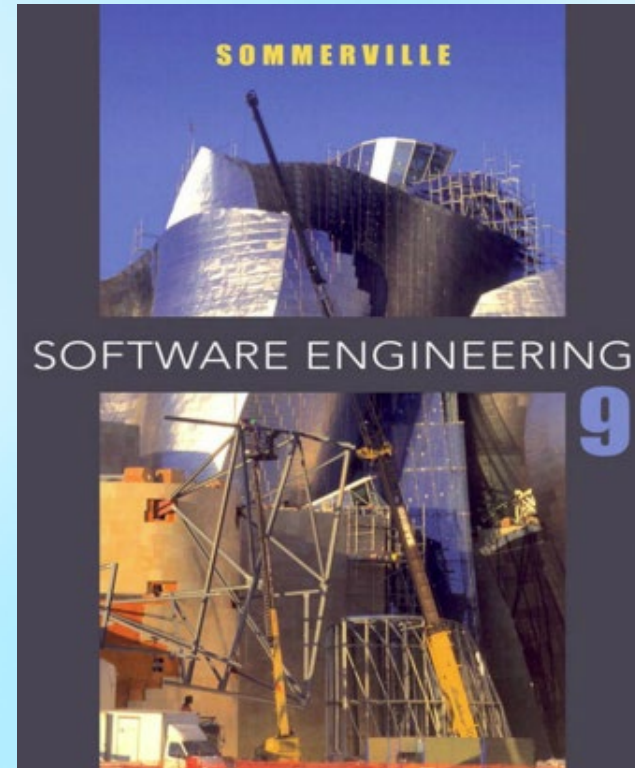
Class Code
byvb74r2



Fall 2025

Book for Reference

- Software Engineering 9th Edition by Ian Sommerville



Mark Distribution



- Mid Semester Exam – 25%
- Final Semester Exam – 40%
- Class Attendance & Performance – 10%
- Course Project – 25%

Is it possible to get A/A+ ?

- Absolutely. Just follow my instructions to the point.

How to fail in this course ?

- Don't follow my instructions.

Instructions



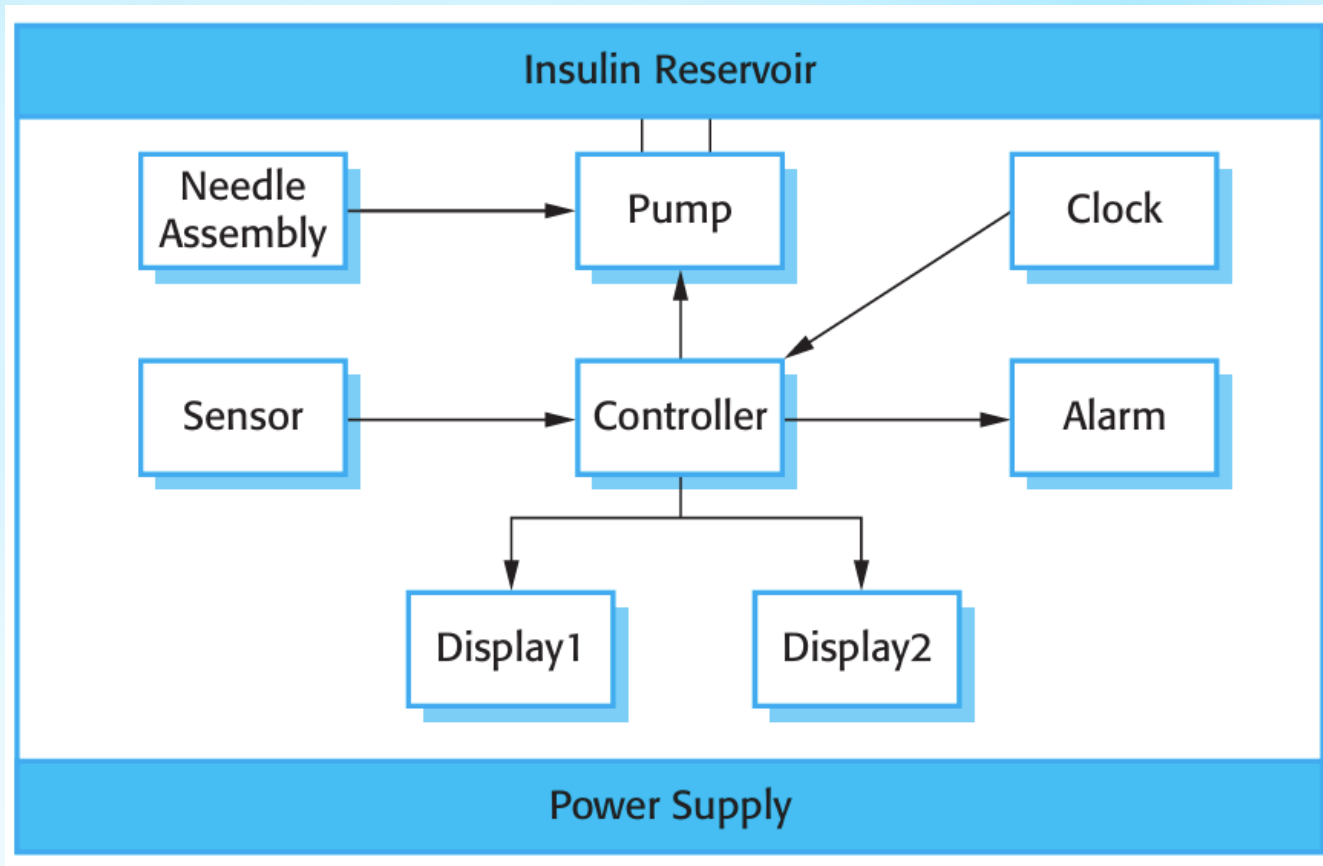
Case studies

- A personal insulin pump
 - An embedded system in an insulin pump used by diabetics to maintain blood glucose control.
- A mental health case patient management system
 - A system used to maintain records of people receiving care for mental health problems.
- A wilderness weather station
 - A data collection system that collects data about weather conditions in remote areas.

Insulin pump control system

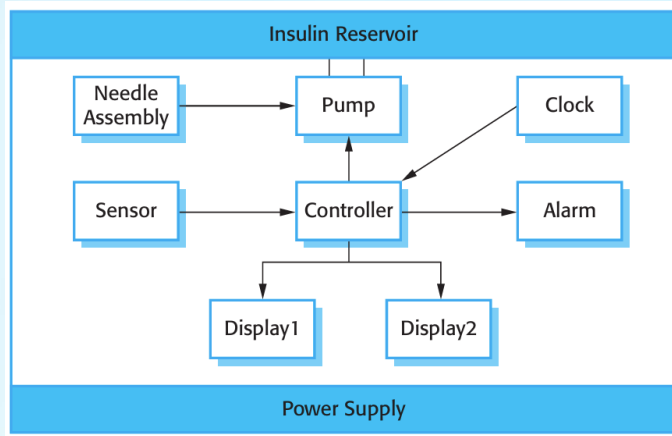
- Collects data from a blood sugar sensor and calculates the amount of insulin required to be injected.
- Calculation based on the rate of change of blood sugar levels.
- Safety-critical system as low blood sugars can lead to brain malfunctioning, coma and death; high-blood sugar levels have long-term consequences such as eye and kidney damage.
- Sends signals to a micro-pump to deliver the correct dose of insulin.

Insulin pump hardware architecture



Insulin pump hardware architecture

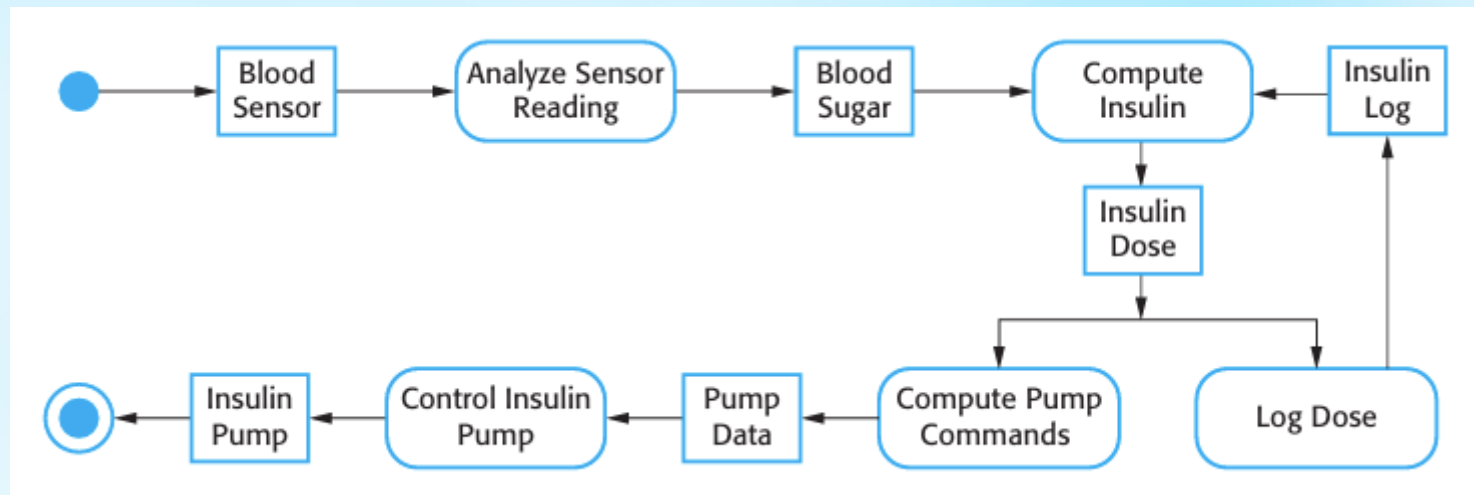
The arrows in this diagram means that there is some communication between these components. The controller is a software component.



In fact, as we shall see later, the system has two displays plus buttons that signal the controller to take certain actions.

The pump delivers a single increment of insulin on receipt of a signal pulse from the controller. Therefore, to deliver 6 insulin increments, 6 pulses are sent to the pump unit. The pump unit includes a pressure sensor so that if the needle assembly is blocked then insulin delivery will be halted

Activity model of the insulin pump



Essential high-level requirements

- Available to deliver insulin when required.
- Perform reliably and deliver the correct amount of insulin to counteract the current level of blood sugar.
- Designed and implemented to ensure that the system always meets these requirements.

A patient information system for mental health care

- A patient information system to support mental health care is a medical information system.
- Most mental health patients do not require dedicated hospital treatment but need to attend specialist clinics regularly.
- These clinics are not just run in hospitals. They may also be held in local medical practices or community centres.

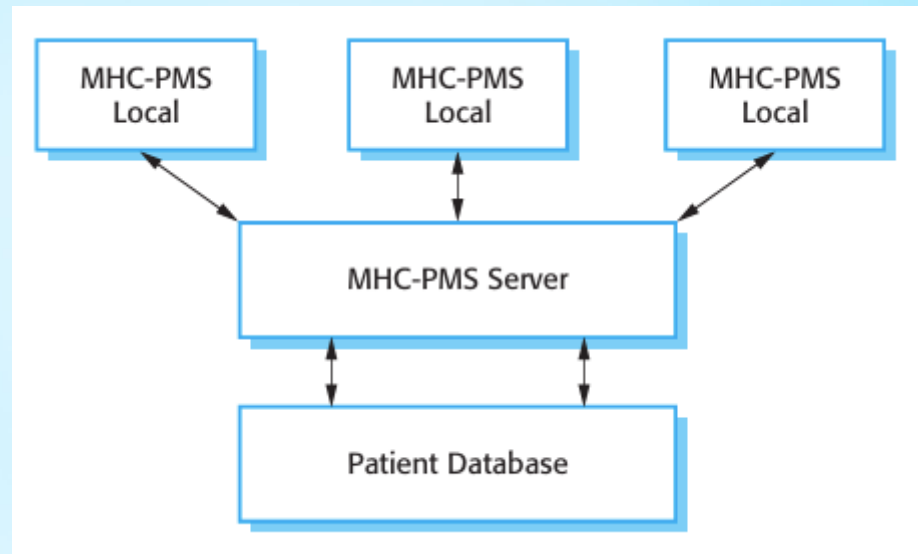
MHC-PMS

- The MHC-PMS (Mental Health Care-Patient Management System) makes use of a centralized database of patient information but has also been designed to run on a PC.
- When the local systems have secure network access, they use patient information in the database but they can download and use local copies of patient records when they are disconnected.

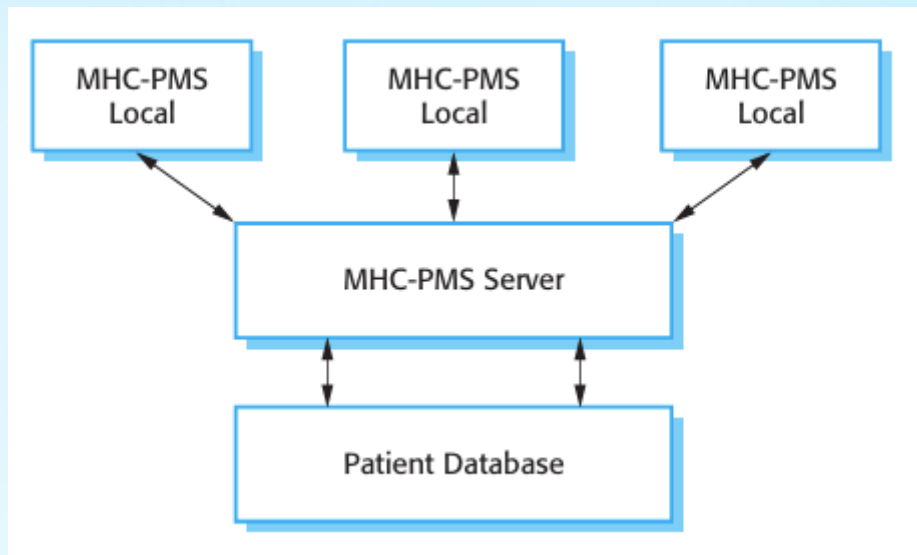
MHC-PMS goals

- To generate management information that allows health service managers to assess performance against local and government targets.
- To provide medical staff with timely information to support the treatment of patients.

The organization of the MHC-PMS



The organization of the MHC-PMS



Patients do not always attend the same clinic but may attend in different hospitals and local health centers.

Patients may be confused and disorganized, miss appointments, deliberately or accidentally lose medication, forget instructions and make unreasonable demands on staff.

Mental health care is safety-critical as a small number of patients may be a danger to themselves and others

MHC-PMS key features

- Individual care management
 - Clinicians can create records for patients, edit the information in the system, view patient history, etc.
- Patient monitoring
 - The system monitors the records of patients that are involved in treatment and issues warnings if possible problems are detected.
- Administrative reporting
 - The system generates regularly management reports.

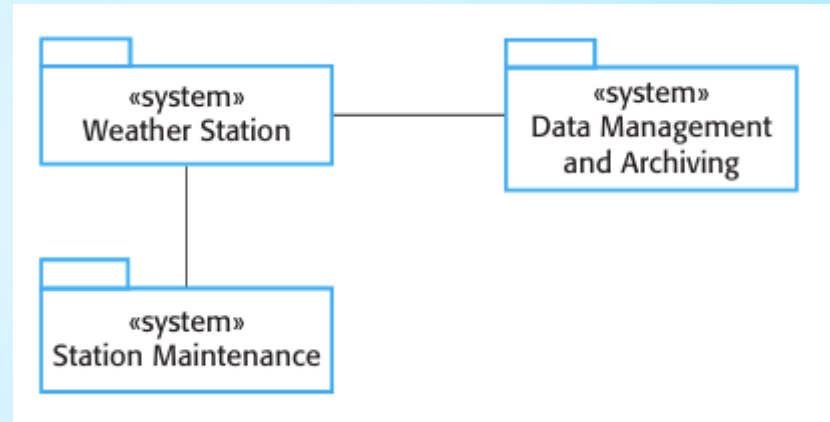
MHC-PMS concerns

- Privacy
 - Patient information is confidential and is never disclosed to anyone apart from authorised medical staff and the patient themselves.
- Safety
 - Some mental illnesses cause patients to become suicidal or a danger to other people.
 - The system must be available when needed otherwise safety may be compromised.

Wilderness weather station

- The government of a country decides to deploy several hundred weather stations in remote areas.
- Weather stations collect data from a set of instruments.

The weather station's environment



Weather information system

- The weather station system
 - Responsible for collecting weather data, carrying out some initial data processing and transmitting it to the data management system.
- The data management and archiving system
 - Collects the data from all of the wilderness weather stations, carries out data processing and analysis and archives the data.
- The station maintenance system
 - Communicates by satellite with all wilderness weather stations to monitor the health of these systems and provide reports of problems.

Additional software functionality

- Monitor the instruments, power and communication hardware and report faults to the management system.
- Manage the system power, ensuring that batteries are charged whenever the environmental conditions permit.
- Support dynamic reconfiguration where parts of the software are replaced with new versions.

Key points

- Software engineers have responsibilities to the engineering profession and society. They should not simply be concerned with technical issues.
- Professional societies publish codes of conduct which set out the standards of behaviour expected of their members.
- Three case studies are used in the book:
 - An embedded insulin pump control system
 - A system for mental health care patient management
 - A wilderness weather station



Thank You

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